Course Name: Basic Electronics Engineering
Course Code: EC-101
Course Type: Institute Core

Contact Hours/Week: 2L **Course Credits:** 02

Course Objectives

- Introduce students to the fundamental concepts of semiconductor devices and their practical applications.
- Enable students to comprehend the operation and applications of transistors.
- Provide students with a basic understanding of JFETs and MOSFETs.

Unit Number	Course Content	Contact Hours
UNIT-01	Diode and Its Applications: PN Junction Diode- Characteristics and Analysis; Types of Diodes- Zener Diode, Photodiodes, LED, Varactor Diode, Rectifiers and Filter Circuit: Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier and their Analysis, L,C and Pi Filters; Series and Shunt Diode Clippers, Clipping at Two Independent Levels, Clamping Operation, Clamping Circuit; Practical Clamping Circuits, Basic Regulator Supply using Zener Diode.	07L
UNIT-02	Bipolar Junction Transistor: Bipolar Junction Transistors: Construction and Characteristics of BJT, Transistor Configuration: CB, CE, CC Configuration; Small Signal Low Frequency Transistor Model (*h*-Parameters), Analysis of Transistor Amplifier using h-parameters.	05L
UNIT-03	Transistor Biasing: Transistor Biasing and Bias Stabilization: Operating Point, Stability Factor, Analysis of Fixed Bias, Collector to Base Bias, Emitter Resistance Bias Circuit and Self Bias Circuit, Bias Compensation Techniques Transistor Switch and Transistor amplifier.	05L
UNIT-04	Field Effect Transistor: Construction and Characteristics of JFET, JFET Biasing Circuit, JFET Amplifier, MOSFET Construction and Characteristics, MOSFET amplifiers.	07L

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Demonstrate a foundational understanding of the operation of various semiconductor devices.

CO2: Explain the working principles of transistors, including different configurations used for designing analog and digital applications.

CO3: Understand the biasing requirements and circuits used for bipolar junction transistors (BJTs) and field-effect transistors (FETs).

CO4: Develop analytical skills to design circuits based on BJT and FET components.

Text Books

1. Integrated Electronics by J. Millman and C.C. Halkias, McGraw Hill Education, India.
2. Electronics Devices and Circuit Theory by R. Boylestad and L. Nashelsky, Pearson India.
3. Electronic principles by L. Malvino, Tata McGraw Hill Education.

Reference Books

1. Microelectronic Circuits by Adel S. Sedra, Kenneth Carless Smith, Oxford University Press.
2. Schaum's Outline of Electric Devices and Circuits by Jimmie J. Cathey, McGraw Hill.
3. Electronic Devices and Circuits by Theodore F. Bogart, Pearson India.